7. 2

2

✓ (2) $$\lim_{n\to\infty} \frac{\sum_{k=1}^{n} \sqrt{k}}{n\sqrt{n}}$$

$$= \int_0^1 \sqrt{x}\, dx$$

✓ (3) $$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{\sqrt{n^2+kn}}$$

$$= \lim_{n\to\infty} \sum_{k=1}^{n} \frac{\frac{1}{n}}{\sqrt{1+\frac{k}{n}}}$$

$$= \int_0^1 \frac{1}{\sqrt{1+x}}\, dx$$

4

$P = \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$ $\quad f(x) = 2 - x^2$

1) ✓ $R(f, P)$ $\quad$ " $\frac{7}{4}$ $\quad$ " 1 $\quad$ " $-\frac{1}{4}$ $\quad$ " -2

$$= \frac{1}{2}\left(f(\tfrac{1}{2}) + f(1) + f(\tfrac{3}{2}) + f(2)\right)$$

$$= \frac{1}{4}$$

2) ✓ $R(f, P)$ $\quad 2$ " $\quad \frac{7}{4}$ " $\quad 1$ " $\quad -\frac{1}{4}$ "

$$= \frac{1}{2}\left(f(0) + f(\tfrac{1}{2}) + f(1) + f(\tfrac{3}{2})\right)$$

$$= \frac{9}{4}$$

3) ✓ $R(f, P)$ $\quad \frac{31}{16}$ " $\quad \frac{23}{16}$ " $\quad \frac{7}{16}$ " $\quad -\frac{17}{16}$ "

$$= \frac{1}{2}\left(f(\tfrac{1}{4}) + f(\tfrac{3}{4}) + f(\tfrac{5}{4}) + f(\tfrac{7}{4})\right)$$

$$= \frac{11}{8}$$

#7 ✓

$\Delta x_k = x_k - x_{k-1}$ 이 되며, $x_0 = 0$, $x_n = b$ 이므로

$$S_n = \sum_1^n f(\xi_k) \cdot \Delta x_k$$

$$= \sum_1^n \frac{x_k + x_{k-1}}{2} \cdot (x_k - x_{k-1})$$

$$= \frac{1}{2} \sum_1^n x_k^2 - x_{k-1}^2$$

$$= \frac{1}{2} (x_n^2 - x_0^2)$$

$$= \frac{1}{2} b^2 \quad //$$

#11 ✓

$$\int_a^{a+b} f(x)\,dx$$

$$= \int_0^b f(x+a)\,dx$$

$x = bt + a$ 라 두면

$dx = b\,dt$

$x = 0 \quad \cdots \quad t = 0$

$x = b \quad \cdots \quad t = 1$

$$\therefore \int_0^1 f(bt+a) \cdot b\,dt$$

$$= \int_0^1 b f(bt+a)\,dt$$

$$= \int_0^1 b f(bx+a)\,dx \quad //$$

7.4

1

(3) ✓ $\int_0^1 \frac{1}{1+x^2}\,dx$

$= \left[\tan^{-1} x\right]_0^1$

$= \frac{\pi}{4}$ //

(8) ✓ $\int_{-1}^{0} x\sqrt{1-x^2}\,dx$

$x^2 = t$

$2x\,dx = dt$

$\frac{1}{2}\int_1^0 \sqrt{1-t}\,dt$

$= \frac{1}{2}\left[\frac{2}{3}(1-t)^{\frac{3}{2}}\right]_1^0$

$= \frac{1}{3}$ f

3

(2) ✓ $\frac{d}{dx}\int_0^{x^2} e^{t^2}\,dt$ // $\frac{d}{dx}\left(\left[G(x)\right]_0^f\right)' = \left(G(f) - G(0)\right)'$

$= e^{x^4} \cdot 2x$ // $= g(f)\cdot f'$

(3) ✓ $\frac{d}{dx}\int_0^{\sqrt{x}} (t+1)^{\frac{1}{2}}\,dt$

$= (\sqrt{x}+1)^{\frac{1}{2}} \cdot \frac{1}{2\sqrt{x}}$ //

4 ✓ $2\int_a^x f(t)\,dt = 2\sin x - 1$

$2\sin a = 1$ $\quad \sin a = \frac{1}{2}$ $\quad \therefore\ a = \frac{\pi}{6}, \frac{5}{6}\pi$

$2f(x) = 2\cos x$

$f(x) = \cos x$, $\quad a = \frac{\pi}{6}, \frac{5}{6}\pi$

5 ✓

$F(1) = 0$

$F'(x) = f(x)$

$F''(x) = f'(x)$

$$f'(t) = \frac{\sqrt{1+t^4}}{t^2} \cdot 2t$$

$$F''(2) = f'(2) = \sqrt{17}$$

11 ✓

(1) $$f(x) = \int_1^{\frac{1}{x}} \sin^3 t \, dt$$

$$f'(x) = \sin^3\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right)$$

$$G\left(\frac{1}{x}\right) - G(1)$$